ASSIGNMENT-4

① (a) $\min \; w^T \Sigma w \quad \text{wrt.} \quad 1^T w = 1.$

So, $L(w, \lambda) = w^T \Sigma w - \lambda (1^T w - 1)$

$\frac{dL}{dw} = 2\Sigma w - \lambda 1 \Rightarrow \frac{dL}{dw} = 0 \Rightarrow 2\Sigma w - \lambda 1 = 0.$

So, $w = \left(\frac{\lambda}{2}\right)\left[\Sigma^{-1} 1\right].$

& $1^T w = 1 \Rightarrow 1^T \left(\frac{\lambda}{2}\right) \Sigma^{-1} 1 = 1 \Rightarrow \left(\frac{\lambda}{2}\right)\underbrace{(1^T \Sigma^{-1} 1)}_{= A} = 1$

let $1^T \Sigma^{-1} 1 = A \Rightarrow \frac{\lambda}{2} A = 1 \Rightarrow \boxed{\lambda = 2/A}.$

$w_{GMV} = \left(\frac{\lambda}{2}\right)\left[\Sigma^{-1} 1\right] \Rightarrow \left(\frac{2}{A}\right)\left(\frac{1}{2}\right)\left[\Sigma^{-1} 1\right]$

$\Rightarrow \left[\frac{\Sigma^{-1} 1}{A}\right]$

$U_{GMV} = w_{GMV}^T \mu \Rightarrow \left(\frac{\Sigma^{-1} 1}{A}\right)^T \mu \Rightarrow \frac{1^T \Sigma^{-1} \mu}{A}$

& let $1^T \Sigma^{-1} \mu = B \Rightarrow \boxed{U_{GMV} = B/A}$

(b) $\sigma^2_{GMV} = w_{GMV}^T \Sigma w_{GMV} \Rightarrow \left(\frac{\Sigma^{-1} 1}{A}\right)^T \Sigma \cdot \left(\frac{\Sigma^{-1} 1}{A}\right).$

$\Rightarrow \frac{1^T \cancel{\Sigma^{-1}} \cancel{\Sigma} \Sigma^{-1} 1}{A^2} \Rightarrow \frac{1^T \Sigma^{-1} 1}{A^2} = \frac{A}{A^2}$

$= [1/A]$

(c) $\mu = \begin{bmatrix} 0.06 \\ 0.10 \\ 0.14 \end{bmatrix}$, $\Sigma = \begin{bmatrix} 0.010 & 0.002 & 0.001 \\ 0.002 & 0.0225 & 0.003 \\ 0.001 & 0.003 & 0.040 \end{bmatrix}$ (2)

Then, Σ^{-1}; (by software pumpy).

$$\Rightarrow \quad \Sigma^{-1} = \begin{bmatrix} 101.96 & -8.81 & -1.89 \\ -8.81 & 48.65 & -3.20 \\ -1.89 & -3.20 & 25.29 \end{bmatrix}$$

$A = 1^T \Sigma^{-1} 1 \Rightarrow 145.088$ | $C = 1.09$

$B = 1^T \Sigma^{-1} \mu \Rightarrow 11.67$ | $D = 22.34$

$w_{GMV} = \dfrac{\Sigma^{-1} 1}{A} \Rightarrow \begin{bmatrix} 0.62894322 \\ 0.2318612 \\ 0.13919558 \end{bmatrix}$

$\mu_{GMV} = \dfrac{B}{A} = 0.0804$

$\sigma_{GMV} = \dfrac{1}{\sqrt{A}} = 0.083$

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(2) (a) $\min \quad w^T \Sigma w \quad$ s.t. $\quad 1^T w = 1 \;,\; \mu^T w = m.$

Lagrangian $(w, \lambda_1, \lambda_2) = w^T \Sigma w - \lambda_1 (1^T w - 1) - \lambda_2 (\mu^T w - m)$

$\dfrac{dL}{dw} = 2\Sigma w - \lambda_1 1 - \lambda_2 \mu = 0.$

$w = \dfrac{1}{2} \Sigma^{-1} (\lambda 1 + \lambda_2 \mu)$

put into both constraints:

$$1^T\left[\frac{1}{2}\Sigma^{-1}(\lambda_1 1 + \lambda_2 4)\right] = 1 \quad ; \quad 4^T\left[\frac{1}{2}\Sigma^{-1}(\lambda_1 1 + \lambda_2 4)\right] = m$$

$$\Rightarrow 1^T\Sigma^{-1}1(\lambda_1) + 1^T\Sigma^{-1}4(\lambda_2) = 2.$$

$$4^T\Sigma^{-1}1(\lambda_1) + 4^T\Sigma^{-1}4(\lambda_2) = 2m$$

$$A = 1^T\Sigma^{-1}1 \quad , B = 1^T\Sigma^{-1}4 \quad , C = 4^T\Sigma^{-1}4 \quad , \Delta = AC - B^2$$

$$A\lambda_1 + B\lambda_2 = 2 \quad \& \quad B\lambda_1 + C\lambda_2 = 2m$$

Solving:

$$\lambda_1 = \frac{2(C - Bm)}{\Delta} \quad , \quad \lambda_2 = \frac{(Am - B)2}{\Delta}.$$

So, $$w^*(m) = \frac{1}{2}\Sigma^{-1}\left(\frac{2(C - Bm)}{\Delta}1 + \frac{2(Am - B)}{\Delta}4\right).$$

$$= \frac{\Sigma^{-1}}{\Delta}\left((C - Bm)1 + (Am - B)4\right).$$

So, $$g = \Sigma^{-1}\left(\frac{A4 - B1}{\Delta}\right) \quad , \quad h = \Sigma^{-1}\left(\frac{C1 - B4}{\Delta}\right).$$

(b) Two Fund Seperation Theorem: Any efficient portfolio (min-var for a given return) can be expressed as linear combo of two distinct efficient portfolios. or

$$w = \alpha w_1 + (1 - \alpha)w_2$$

as we have $w(\mu) = \mu g + h$ which is linear equation so any other efficient must also lie on it and we can make a line using any two points on it.

$$w(m) = \alpha w^*(m_1) + (1-\alpha) w^*(m_2)$$

$$= \alpha(m_1 g + b) + (1-\alpha)(m_2 g + b).$$

$$= (\alpha + (1-\alpha)) b + [\alpha m_1 g + (1-\alpha) m_2 g]$$

$$m_0 g + b = [\alpha m_1 + (1-\alpha) m_2] g + b.$$

$$m_0 g = (\alpha m_1 + (1-\alpha) m_2) g.$$

$$\hookrightarrow \boxed{\alpha = \frac{m_0 - m_2}{m_1 - m_2}}$$

Done.

(c) Let: $m = 0.1$ and then we need to find.

A, B, C, D and σ.

$$\hookrightarrow A = 145.088 \quad , B = 11.67 \quad , C = 1.09.$$

$$D = 22.3353$$

$$w(m=0.1) = \begin{bmatrix} 0.32786885 \\ 0.3442623 \\ 0.32786885 \end{bmatrix}$$

Variance ($\mu = 10\%$) $= 0.009$

$$\sigma(\mu = 10\%) = 0.0969.$$

(3) (a)

Minimize

S.t.

$L(w$

Stationarity

Primal F

Dual

Complem

(b) If:

Says

or Port

(3) (a) Full KKT for long only positions:

$$\text{Minimize} : \quad w^T \Sigma w.$$

$$\text{S.t.} : \quad 1^T w = 1.$$

$$\mu^T w = m.$$

$$w_i \geq 0.$$

$$\mathcal{L}(w, \lambda_1, \lambda_2, \lambda_3) = w^T \Sigma w - \lambda_1 (1^T w - 1) - \lambda_2 (\mu^T w - m) - \lambda_3^T w$$

$$\lambda \geq [\Sigma(\lambda_3); w_i].$$

Stationarity:

$$2\Sigma w - \lambda 1 - \lambda_2 \mu - \lambda_3 = 0.$$

Primal Feasibility:

$$1^T w = 1$$

$$\mu^T w = m$$

$$w_i \geq 0.$$

Dual Feasibility:

$$\hookrightarrow \quad (\lambda_3)_i \geq 0.$$

Complementary Slackness:

$$\hookrightarrow \quad (\lambda_3)_i \, w_i = 0.$$

(b) If: $\lambda_i > 0$; Then, $w_i = 0$.

Says that Asset lies on boundary.

Or Portfolio touches constraint edge.

Or also.

$(d_3)_i > 0$ says $w_i = 0$ and constraint $w_i \geq 0$ is active (6)

means optimal portfolio excludes asset i as including it would worsen variance with tuned constraint.

(c) $\mu_1 = 0.06$ $\sigma_1 = 0.1$ $\rho = 0.3$

$\mu_2 = 0.12$ $\sigma_2 = 0.18$ $m = 0.08$

$$\Sigma = \begin{bmatrix} 0.01 & 0.0054 \\ 0.0054 & 0.0324 \end{bmatrix}$$

$w_1 + w_2 = 1$, $w_i \geq 0$

$\mu_1 w_1 + w_2 \mu_2 = m$.

$0.06\, w_1 + 0.12\, w_2 = 0.08$ & $w_2 = 1 - w_1$

$0.06\, w_1 + 0.12(1 - w_1) = 0.08$

$\hookrightarrow$ $w_1 = 2/3$, $w_2 = 1/3$.

And all $w_i \geq 0$ $\Rightarrow$ Satisfied conditions.

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(4) $K = 2$ from $\{A_1, A_2, A_3\}$ to minimize variance &

$1^T w = 1$, $\mu^T w \geq 9\%$, $w_i \geq 0$.

(4) MIQP Constraints Formulation :

$\min \quad w^T \Sigma w$

s.t. $1^T w = 1$

$\mu^T w \geq 0.09$

Big - M Co

Cardinality :

OBJECTIVE

$1^T w$

(b) All

↳ For

↳ For

↳ for [

$w_i \geq 0$

$z_i \in \{0,1\}$ $\Rightarrow$ Binary constraint for inclusion-exclusion.

& $\sum z_i \leq 2$

Big-M Constraint: $w_i \leq M z_i$

iff $z_i = 0 \Rightarrow w_i = 0$.

Cardinality: $\sum z_i \leq 2$

OBJECTIVE: $\min w^T \Sigma w$ s.t.

$1^T w = 1$, $\mu^T w \geq 0.09$, $w_i \geq 0$.

$w_i \leq z_i$, $\sum z_i \leq 2$.

(b) All 3C2 pairs can be $[A_1 A_2 \mid A_2 A_3 \mid A_1 A_3]$

↳ For $[A_1, A_2]$ $\Rightarrow$ $w = [0.719, 0.281]$

Return = 0.071

Std = 0.088 $\Rightarrow$ Not Possible.

↳ For $[A_1 A_3]$ $\Rightarrow$ $w = [0.8125, 0.1875]$

Return = 0.075

Std = 0.0911 $\Rightarrow$ Not possible.

↳ for $[A_2, A_3]$ $\Rightarrow$ $w = [0.6549, 0.3451]$

Return = 0.1138

$JH_d \Rightarrow 0.1255$

↳ Feasible + Optimal pair.

↳ All the $\bar{Z}_{12}$ $\bar{Z}_{23}$ $\bar{Z}_{13}$ can be found by taking respective rows and truncating the extra col not corresponding to rows

(b)

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(5) $(w_1, w_2) = (0.6, 0.4)$, $T = 1\%$.

r_{1t}	3	1	-2	4	-3	2
r_{2t}	5	-1	2	-3	1	4

$R_t = 0.6\, r_{1t} + 0.4\, r_{2t}$.

(4) $R_1 = 1.8 + 2.0 = 3.8$

$R_2 = 0.6 - 0.4 = 0.2$

$R_3 = -1.2 + 0.8 = -0.4$

$R_4 = 2.4 - 1.2 = 1.2$

$R_5 = -1.8 + 0.4 = -1.4$

$R_6 = 1.2 + 1.6 = 2.8$

$JV = \frac{1}{T} \sum \min(0, R_t - r)^2$

$= \frac{1}{6}\left[2.8^2 + 0^2 + 0^2 + 0.2^2 + 0^2 + 1.8^2\right]$

$= \frac{1}{6}\left[7.84 + 0.04 + 3.24\right]$

$= \frac{1}{6} \sum \left[0^2 + 0.8^2 + 1.4^2 + 0^2 + 2.4^2 + 0^2\right]$

$\Rightarrow \frac{1}{6}[\ 0.64 + 1.96 + 5.76\ ]$

$\Rightarrow$ 1.3933 % percentage square.

obs: $\left[\frac{1.3933}{(100)^2}\right]$.

(b) let $d_t = \min(R_{pt} - T, 0)$.

Semivariance $\Rightarrow \frac{1}{T}\sum\left(\min[0, r_t^T w - T]^2\right)$

So, $d_t \leq 0$. & $d_t \leq R_{pt} - T$

$$\left[\begin{array}{lll} \min & \frac{1}{T}\sum_{t=1}^{T} d_t^2 & \\ \text{s.t.} & 1^T w = 1 & d_t \leq R_{pt} - T \\ & \mu^T w \geq m. & d_t \leq 0. \\ & & w_i \geq 0. \end{array}\right]$$

Converting $d_t = -s_t$. all constraint ≥ 0.

$$\begin{array}{lll} \min & \frac{1}{T}\sum s_t^2 & \\ \text{s.t.} & 1^T w = 1 & \\ & \mu^T w \geq m & \\ & s_t \geq T - r_t^T w & \\ & s_t \geq 0 & \\ & w_i \geq 0 & \end{array}$$

$$R = [r_1^T \; r_2^T \; \dots \; r_T^T]$$

$$Rw = [r_1^T w \; r_2^T w \; \dots \; r_T^T w]$$

$$\min_{w,s} \quad \frac{1}{T} s^T s = f.$$

$$\text{s.t.} \quad 1^T w = 1 \quad , \quad s \geq \tau 1 - Rw.$$

$$u^T w \geq m \quad , \quad s \geq 0.$$

$$w \geq 0$$

$$\nabla f(s) = \frac{2}{T} s. \rightarrow \text{GRADIENT}$$

$$\nabla^2 f(s) = \frac{2}{T} I \rightarrow \text{HESSIAN.}$$

& $I > 0 \Rightarrow \frac{2I}{T} > 0.$

↳ So, $f(s)$ is strictly convex.

⇒ EQUALITY: $[1^T w = 1] \Rightarrow$ Affine

↳ And affines are convex.

⇒ INEQUALITY: $s \geq \tau 1 - Rw \Rightarrow Rw + s > \tau 1$

↳ This is linear and linear inequality are convex.

⇒ Non-Negativity: $s \geq 0 \quad , \quad w \geq 0.$

↳ Half Spaces ⇒ CONVEX.

c) All constraints are linear or affine.

So, Feasible set is convex.

↳ So, Semivariance minimization is a convex quad. program.

(c) $SV(w, \tau) \leq Var(R_p)$ when $\tau = \mu_p$.

$Var(R_p) = E[(R_p - \mu_p)^2] \Rightarrow E[(R_p - \mu_p)^2 \cdot 1_{R_p \geq \mu_p}] +$

$E[(R_p - \mu_p)^2 \cdot 1_{R_p < \mu_p}]$

First term ≥ 0 & Second term $= SV$.

So, $Var(R_p) = E[(R_p - \mu_p)^2 \cdot 1_{R_p \geq \mu_p}] + SV(w, \tau_{\mu_p})$.

So, $Var(R_p) \geq SV(w, \mu_p)$. $\rightarrow \geq 0$.

Second moment always

Also, $\min(0, x^2) \leq x^2$

↳ Taking Expectation on both.

$SV(w, \mu_p) \leq Var(R_p)$.

↳ $Var(R_p) = E[(R_p - \mu_p)^2]$.

$SV(w, \mu_p) = E[\min(0, R_p - \mu_p)^2]$

Let $x = R_p - \mu_p$ & $E[x] = 0$.

$Var(R_p) = E[x^2]$

$SV(w, \mu_p) = E[\min(0, x)^2]$

(12)

$\min(0,x^2)^2 \leq x^2$.

C1: $x \geq 0$ $\quad \min(0,x) = 0 \Rightarrow \min(0,x)^2 = 0$.

& $0 \leq x^2$

C2: $x \leq 0$ $\quad \min(0,x) = x \Rightarrow \min(0,x)^2 = x^2$.

& $\min(0,x)^2 \doteq x^2$

& $A \leq B \Rightarrow E[A] \leq E[B]$ [For all cases or outcomes]

$\therefore$ $Var(W,U_p) \leq Var(R_p)$.

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(6) (a) $VaR_\alpha(L) = \inf\{x \in R : P(L \leq x) \geq \alpha\}$.

α- quantile of loss distribution.

It is an point on CDF curve. what probability is same as confidence level.

(b) $\mu_p = 0.05\%/day$; $\sigma_p = 2\%/day$; VAL = 1×10^6

(losses =) $-2, -1.3, -0.2, 0.5, 0.8, 1.1, 1.8, 2.1, 3.5, 4.2$

$Var(90) \Rightarrow$ Step = 0.9×10

= 9^{th} Value

$\therefore$ $\hat{Var}_{90} = 3.5\%$

(7) (9) CVaR$_\alpha$(

&

Lo Th

on

(5) Fo $\alpha = 9$

closes

CVaR

Fo $\alpha = 80$

& los

CV$_\alpha$R

& yes

(c) $\mu_p = 0$

(V

(a) (g) $CVaR_\alpha(L) = E[L \mid L \geq VaR_\alpha(L)]$

$$= \frac{1}{1-\alpha} \int_{VaR_\alpha}^{\infty} x \, dF_L(x)$$

↳ This basically a conditional + combinational probability on the VaR_α itself.

(b) For $\alpha = 90\%$, $VaR = 3.5$.

losses $\geq 3.5 \Rightarrow \{3.5, 4.2\}$

$CVaR = (3.5 + 4.2)/2 = 3.85\%$.

For $\alpha = 80\%$, $VaR = 2.1$

& losses $\geq 2.1 \rightarrow (2.1, 3.5, 4.2)$.

$CVaR = (2.1 + 3.5 + 4.2)/3 = 3.267$.

& yes $CVaR \geq VaR$ in both.

(c) $\mu_p = 0$, $\sigma_p = 2\%$, $[\Phi(2.326) = 0.02667]$

$$CVaR = \frac{\phi(z)}{1-\alpha} \sigma$$

$$= \frac{0.02667}{0.01} \times 0.02$$

$$= 0.0533$$

$$\Rightarrow 5.33\%$$

(d) CVaR = Avg. of All tail losses.

VaR = minimum tail losses.

And is true that?

$$\min(\text{Lost}) \leq \text{Avg}(\text{Lost}).$$

Let L be a loss fun with CDF F_L & for any confidence level α in $(0,1)$:

$$(\text{VaR}_\alpha(L) \geq \text{VaR}_\alpha(L).$$

$\Rightarrow$ For any probability level: $u \geq \alpha$.

$$\text{VaR}_u(L) \geq \text{VaR}_\alpha(L). \qquad \forall u \in [0,1].$$

↳ Quantile fun is non-decreasing in probability level.

$$\Rightarrow (\text{VaR}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 \text{VaR}_u(L)\,du.$$

$$\overset{!!}{\Rightarrow} \int_\alpha^1 \text{VaR}_u(L)\,du \geq \int_\alpha^1 \text{VaR}_\alpha(L)\,du.$$

$$\geq \text{VaR}_\alpha(L)(1-\alpha).$$

$$(\text{VaR}_\alpha(L) \geq \left(\frac{1}{1-\alpha}\right)\text{VaR}_\alpha(L)(1-\alpha)$$

$$\geq \text{VaR}_\alpha(L).$$